Dheepak Karan Elumalai Santhakumari

ECE GRAD STUDENT · AI AGENTS, LLMs & CLOUD

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Software engineer with 1+ year of production experience, now building AI agents, RAG pipelines, and LLMOps tools. Resilient and calm under pressure, takes full ownership, learns fast, and delivers reliable results on tight budgets and timelines.

01 EDUCATION

Northeastern University — MS Electrical & Computer Engineering · GPA 3.9 — Boston · Expected May 2028
Computer Vision, ML & Algorithms · *Coursework: Machine Learning, Large Language Models, MLOps*

Anna University — BE Electrical & Electronics Engineering — India · 2022

02 TECHNICAL SKILLS

LANGUAGES & BACKEND	Python, SQL, Java, Bash, FastAPI, REST APIs, Spring Boot, PostgreSQL, Redis, Apache Kafka
GENAI & AGENTS	LangGraph, MCP, multi-agent systems, tool calling, prompt engineering, guardrails, human-in-the-loop
RAG & EVALS	Embeddings, Sentence-Transformers, Chroma, Qdrant, semantic search, LLM evaluation, recall@k, MRR
LLMS & ML	PyTorch, Hugging Face Transformers, LoRA / QLoRA, quantization, Ollama, CUDA, scikit-learn, XGBoost, pandas
LLMOPS & CLOUD	AWS (Lambda, DynamoDB, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Git, Linux

03 PROJECTS

AI Incident Agent | `incident-triage-agent` — LangGraph · MCP · Chroma · Pydantic

- Built two tool-calling AI agents (investigator + reviewer) that diagnose outages from metrics, logs, and traces, tested on RCAEval.
- Added guardrails and human-in-the-loop approval for risky fixes that blocked 7 of 7 attacks, including prompt injection.

LLMOps Gateway | `llm-gateway` — AWS Lambda · DynamoDB · Terraform · CloudWatch

- Built an OpenAI-compatible LLM gateway that suspends runaway agents (loops, retry storms, token bombs) within 5 seconds.
- Deployed serverless on the AWS free tier at $0/month, catching 12 of 12 runaway test runs with 0 false blocks on normal traffic.

RAG Search | `semantic-rag-search` — Sentence-Transformers · Ollama · GitHub Actions

- Built a RAG pipeline that embeds notes for semantic search and answers with citations, with Gemini → OpenAI → Ollama fallback.
- Added retrieval evals (recall@4, MRR) as a CI gate and reached 4.6 ms p50 search over 5,000 chunks without a vector DB.

NLP Sports Analytics | `xg-from-words` — Qdrant · MiniLM · pytest

- Built an expected-goals model that rates shots from free commentary text at 0.78 AUC vs 0.81 for a paid coordinate-data model.
- Fixed label leaks that faked a perfect 1.0 AUC and showed the model holds 0.77–0.79 AUC on 5 other leagues without retraining.

04 EXPERIENCE

Northeastern University — Student & Instructional Assistant (part-time) — Boston · 2026 — Present

- Developed a personal AI **agent** that answers questions about Northeastern university resources.
- Building PAWS, a Microsoft Copilot Studio tracker agent that balances academic deadlines and job-search tasks in one place.
- Created **custom Claude skills** and a personal research agent that turn course topics into visual explanations for faster study.
- Built a daily LeetCode bot (GitHub Actions cron, Telegram) that picks non-repeating problems and tracks completion progress.
- As a part-time Instructional Assistant, resolve live hybrid-class issues (Teams, classroom AV) for faculty through ServiceNow.

Guardian Life — Software Engineer (Fortune 250 U.S. Insurance Company) — India · 2022 — 2023

- Collaborated in an Agile Scrum team building microservices for an AWS migration, syncing remotely with U.S. analysts.
- Handled testing and Jenkins deployment of releases from shared Linux servers to a cluster, backed by 120+ automated tests.
- Wrote a monitoring script that caught **50%+ of production issues** before users did, and triaged 15+ incidents to root cause.
- Made a policy lookup API **~20x faster** by fixing an inefficient database query pattern and adding caching.
- Supported data scientists with SQL data pulls, data prep, and EDA for their ML models (XGBoost).

05 TEACHING & ACHIEVEMENTS

TEACHING	Volunteer programming instructor for a community initiative · Class leader at Anna University, mentoring juniors.
ACHIEVEMENT	Team rank 51 of 250 (top 21%) in a Kaggle LLM fine-tuning contest, running parallel fp16 inference on 2 GPUs.